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draft_final_report_25_26J_148_IT22141606 (1)



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AUTOMATIC CRICKET VIDEO SEGMENTATION BY PLAYER IDENTITY AND EVENT TYPE

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April 2026

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9

Dissertation submitted in partial fulfilment of the requirements for the Bachelor of
Science Special Honors Degree in Information Technology

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Department of Information Technology

Sri Lanka Institute of Information Technology

Malabe, Sri Lanka

April 2026

DECLARATION

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, Prof. Anuradha Karunasena, for the continuous guidance, encouragement, and academic insight provided throughout this research. I am also grateful to my co-supervisor, Ms. Adya Dissanayake, for her valuable suggestions and support during the design, implementation, and refinement of the system. My appreciation is extended to the Department of Information Technology, Sri Lanka Institute of Information Technology, for providing the academic environment and resources required to complete this work. I would also like to thank all those who contributed to the collection of ideas, technical discussions, and constructive feedback that helped improve this study.

Finally, I would like to express my gratitude to my team members who were hardworking and supportive throughout the entire project to make this possible. Most importantly, I would appreciate the continuous backup offered by my parents. I am more than grateful for this work being an opportunity to showcase my potential.

ABSTRACT

Cricket match videos contain valuable information for performance analysis, coaching, and fan engagement, yet accessing player-specific and event-specific moments in full-length recordings usually requires manual review. This study presents an automated cricket video segmentation and retrieval system that combines scoreboard text extraction, umpire-signal-based event detection, player identification, timestamp synchronization, and semantic retrieval. Scoreboard information is extracted from broadcast frames using EasyOCR and then cleaned to identify contextual match information. A two-stage event detection pipeline uses EfficientNet-B0 to detect umpire frames and ResNet-18 to classify major match events such as FOUR, SIX, and WICKET. Named Entity Recognition, fuzzy roster matching, and post-processing are used to associate detected events with batsmen and bowlers. The system further enriches outputs through ordinal event labeling, wicket attribution, over segmentation, transcript-assisted exploration, and MiniLM-FAISS based semantic search. A web-based interface allows users to upload match videos, browse structured outputs, and navigate directly to relevant timestamps. Experimental results show an accuracy of 0.91 for umpire detection and 0.83 for pose-based event classification, demonstrating the practical value of the proposed multimodal approach. The final system reduces manual effort and improves retrieval efficiency for analysts, coaches, and cricket fans.

Keywords: Cricket Video Analysis, Player Identification, Event Detection, Semantic Search, Sports Video Analytics

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Programming Interface
FAISS	Facebook AI Similarity Search
NER	Named Entity Recognition
NLP	Natural Language Processing
OCR	Optical Character Recognition
STT	Speech-to-Text
UI	User Interface

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1. INTRODUCTION

1.1 Background and Literature Survey

Cricket is one of the most widely followed sports in the world and generates an enormous volume of broadcast video content through international tournaments, franchise leagues, bilateral series, and domestic competitions. These video recordings are valuable resources for performance analysis, tactical decision-making, coaching support, talent development, media production, and fan engagement. Coaches and analysts often need to review match situations in detail to understand batting patterns, bowling effectiveness, wicket-taking moments, and tactical sequences. Similarly, fans increasingly expect fast access to memorable moments involving their favorite players rather than manually browsing hours of video footage. However, despite the abundance of cricket video content, identifying and retrieving specific moments from full-length match recordings remains a difficult and time-consuming task. In many practical situations, users must still manually inspect lengthy recordings to locate a specific batsman, bowler, boundary, wicket, or sequence of match events. This manual review process remains a major practical limitation in cricket video analysis and retrieval.

The broader field of sports video analysis has advanced considerably over the last two decades. Earlier research in this area focused on extracting semantic meaning from sports videos using handcrafted visual features, rule-based event detection, and manually designed heuristics. These traditional approaches were useful for detecting broad categories of events and for building early highlight-generation systems, but they had limited scalability and were often sensitive to variations in camera angles, lighting, and broadcast conditions. Over time, developments in machine learning and deep learning made it possible to replace many handcrafted pipelines with more robust data-driven techniques. Recent work in sports video understanding increasingly relies on convolutional neural networks, deep feature learning, multimodal fusion, and embedding-based retrieval methods to improve the quality of event recognition and video indexing

[1], [2], [5], [6]. This shift is important because sports videos are inherently complex, involving visual, textual, and contextual cues that need to be interpreted together rather than in isolation.

In cricket specifically, many previous studies and practical systems have focused primarily on event detection rather than player-centered retrieval. A large amount of work has been dedicated to detecting boundaries, wickets, deliveries, or highlight-worthy match events. Such systems are useful because they reduce the manual effort required to find important match moments. However, they often remain limited to isolated event spotting or summary generation. In other words, they can often indicate that a wicket or a FOUR has occurred, but they do not always provide a complete operational framework that links the event to the correct player identities, contextual scoreboard information, precise timestamps, and a searchable retrieval structure for full-length videos. This leaves an important gap between automatic event recognition and practical user-oriented exploration of full cricket matches. A real user may want to search for all wickets involving a certain bowler, all boundaries by a specific batsman, or the second wicket in the innings, but many existing systems are not built to support such structured queries across a complete match recording.

Another major research direction in sports video understanding is the use of embedded text and contextual overlays. Broadcast sports videos often contain scoreboard overlays that provide rich semantic information such as team score, wickets, overs, player names, and match context. In cricket, this scoreboard information is especially useful because it changes dynamically with the progress of the match and frequently displays the batsmen and bowler currently involved in play. Optical Character Recognition (OCR) methods make it possible to extract this information directly from video frames. EasyOCR is used for this purpose because scoreboard extraction provides an effective way to recover player-related context from cricket broadcasts. This is important because player appearance alone may not always be sufficient for reliable identification, especially when camera distance, motion blur, or broadcast editing reduce visual clarity. By using

scoreboard OCR, the system can access contextual information that is directly produced by the broadcast itself [7], [9].

Alongside scoreboard extraction, the system incorporates computer-vision-based event detection using umpire signals. This is a particularly strong design choice because umpire gestures in cricket provide a clear and structured visual indicator of important match events such as FOUR, SIX, and wicket decisions. Rather than relying only on general action recognition, the system uses a two-stage event detection pipeline in which an EfficientNet-B0 model first identifies whether an umpire is present in a frame, and then a ResNet-18 model classifies the specific gesture into an event category. This design improves robustness by filtering irrelevant frames before event classification is applied. It demonstrates how domain-specific visual structure can be used to build a more reliable cricket analysis pipeline [10], [11]. The use of umpire-signal recognition also highlights an important characteristic of cricket broadcasts: context does not come only from scoreboard overlays, but also from the formal visual language of umpire gestures.

A further important dimension of recent research is multimodal learning, which combines visual, textual, and contextual information to improve video understanding. This is especially relevant in cricket because no single modality is always reliable on its own. OCR may suffer from broadcast noise, visual event detection may be affected by camera angle changes, and isolated textual evidence may not fully capture match context unless it is aligned with time. To address this challenge, the overall framework integrates multiple components into a unified pipeline, including scoreboard text extraction, event detection, Named Entity Recognition (NER)-based player identification, timestamp synchronization, and semantic indexing. This combination enables the system to associate events and players more reliably than a single-modality approach. Player identification is performed using NER applied to OCR-extracted text, and the resulting player entities are synchronized with detected match events using timestamps. This multimodal design is a major contribution because it reflects the real structure of cricket broadcasts, where meaning is distributed across text overlays, visual cues, and temporal context rather than being contained in one source alone [8], [12].

Another significant advance is the move from raw event detection toward searchable semantic retrieval. Traditional highlight-generation systems usually produce clips or event markers, but they do not always support flexible natural-language-style exploration. In contrast, the framework converts extracted event descriptions and contextual records into MiniLM sentence embeddings and stores them in a FAISS-based similarity index. This allows the final system to support semantic search over processed cricket match content. In practical terms, this means that users are not limited to browsing a timeline manually; instead, they can search for player appearances, event types, wicket-related sequences, and other contextual cricket moments in a more efficient manner. This retrieval layer transforms detected outputs into a searchable multimedia knowledge structure, greatly improving accessibility for coaches, analysts, and fans [3], [4].

The completed framework also shows a meaningful evolution from the earlier proposal stage. The initial direction placed stronger emphasis on scoreboard OCR and commentary-related analysis, whereas the final system moved toward a more implementation-ready and cricket-specific end-to-end architecture. In the final design, scoreboard OCR remains central, but it is strengthened by umpire-signal-based event detection, roster-aware player association, structured timestamp linking, over-based browsing support, wicket attribution, transcript-assisted exploration, and semantic search. This evolution reflects a deeper understanding of what makes a cricket video segmentation system practically useful after implementation and evaluation. It also shows that the final solution is not only technically richer than the original proposal, but also more aligned with the needs of real users who want structured access to match moments rather than only isolated detections.

Therefore, this study contributes a practical end-to-end framework for automatic cricket video segmentation by player identity and event type. The framework combines OCR-based scoreboard extraction, computer-vision-based event recognition, NER-supported player identification, timestamp synchronization, and semantic retrieval inside a web-based platform. Instead of requiring users to manually review long cricket recordings, the system generates structured event records containing event type, timestamp, player information, and searchable descriptions. This makes the final output more useful for

downstream exploration, analysis, and retrieval. In this way, the study extends prior sports video literature by shifting from isolated event spotting toward a more complete and operational multimedia retrieval system for cricket.

1.2 Research Gap

Although existing research in sports video analysis has achieved notable progress, important gaps remain when the goal is to build a practical, full-length cricket video retrieval system. One major issue is that many systems still depend heavily on manual video review. Even when event detection models can identify certain match events automatically, users often still need to inspect the results manually to determine which players were involved, when exactly the event occurred, and whether it is relevant to their analysis goal. This problem becomes more serious in cricket because match videos are often several hours long and contain repeated visual patterns, multiple camera cuts, and a large volume of contextual information. This manual dependency remains a major weakness in current workflows for player-specific and event-specific retrieval.

A second gap is that many existing solutions are primarily event-centric rather than player-centric. They are designed to detect key events such as boundaries or wickets, but do not always support comprehensive player-based filtering across full match recordings. For many practical users, however, event type alone is not sufficient. Coaches may want all events involving a particular bowler, fans may want all boundaries scored by a specific batsman, and analysts may want to review sequences involving a certain player across an innings. These use cases require event–player linkage, not just event detection. The generation of structured records that connect detected events with batsmen, bowlers, and timestamps directly addresses this unresolved gap in prior systems, which often stop at event identification without building a player-aware retrieval layer.

A third limitation is the lack of integrated multimodal automation. Existing systems often specialize in one task only, such as OCR extraction, highlight generation, action

recognition, or semantic indexing. Very few operational platforms combine scoreboard extraction, visual event classification, player identification, structured timestamp synchronization, over-level reasoning, and semantic search in a single working pipeline. As a result, many available methods remain fragmented. They can solve a narrow task, but they do not offer a complete end-to-end retrieval experience for real users working with full cricket recordings.

Another gap is the limited availability of richer contextual retrieval. In many prior systems, outputs are represented only as isolated highlight clips or event markers. This restricts the usefulness of the system because real analysis often depends on richer structure. For example, a useful cricket retrieval system should ideally support not only event labels, but also timestamped event–player association, ordinal context such as first wicket or second four, wicket attribution, over boundaries, transcript support, and semantic query handling. The inclusion of post-processing modules such as ordinal enrichment, wicket attribution, and over segmentation, together with player-wise and event-wise navigation, shows that current systems still leave significant room for improvement in contextual richness and structured exploration.

There is also a practical research gap between offline model performance and usable application deployment. Some prior work demonstrates strong model accuracy on narrow tasks, but does not integrate the models into a platform that users can employ for interactive retrieval. The final framework addresses this by embedding the outputs inside a web-based platform that supports player filtering, event navigation, over-based browsing, transcript-assisted exploration, wicket attribution display, and semantic search with direct video playback links. This integration is important because a model becomes truly useful in practice only when its results can be accessed, interpreted, and navigated efficiently by end users.

Therefore, the overall research gap can be summarized as follows: despite advances in sports video analysis, there is still a lack of a practical end-to-end system for full-length cricket broadcasts that can automatically combine player identification, event detection,

timestamped association, contextual structuring, and semantic retrieval within one platform. This gap justifies the need for a more complete cricket video understanding system capable of reducing manual effort while generating searchable and player-aware match segments.

1.3 Research Problem

Cricket videos usually extend over a long duration, often spanning several hours. This creates a major challenge for analysts, coaches, and fans who need to locate moments featuring a particular player or a specific type of match event. Manually searching through such recordings is inefficient, repetitive, and impractical, especially when multiple matches need to be reviewed. A coach may want to inspect all wicket-related moments involving a certain bowler, an analyst may want to analyze a batsman's scoring patterns, and a fan may simply want to find memorable player-specific highlights. In each of these cases, the lack of direct structured access to relevant video segments creates a clear usability problem.

Although deep learning has significantly improved sports video analysis, current approaches are still limited when it comes to building a unified, practical pipeline for cricket. A useful cricket retrieval system must do more than detect isolated events. It must also identify the players involved, connect these players to the detected events, synchronize the results using timestamps, and expose the outputs through an interface that supports fast exploration and retrieval. In cricket broadcasts, this requires the intelligent use of multiple information sources. Scoreboard overlays contain player and match-state information, umpire gestures provide strong visual signals for event classification, and text-processing methods can help normalize and interpret the extracted context. Without integrating these signals into one pipeline, the outputs remain fragmented and less useful for real-world analysis.

Accordingly, the research **problem addressed in this study is the** absence of a practical end-to-end multimodal system that can automatically connect player identity, event type, and timestamped navigation for full-length cricket videos. Existing systems may detect events or generate highlights, but they generally do not provide a reliable framework that links who was involved, what occurred, and when it happened in a way that supports structured retrieval. This missing integration limits both research progress and practical usability.

The research problem can therefore be stated more formally as follows:

How can a multimodal cricket video analysis framework be designed to automatically detect important match events, identify the corresponding players, synchronize them through timestamps, and transform the outputs into structured, searchable video segments for interactive retrieval from full-length cricket recordings?

This problem motivates the development of a retrieval-oriented architecture that combines OCR-based scoreboard extraction, umpire-signal-based event recognition, NER-supported player association, timestamp synchronization, semantic indexing, and a web-based interface. By solving this problem, the study aims to reduce manual review effort and improve access to important cricket moments for analysts, coaches, and fans.

1.4 Research Objectives

Main Objective

To develop an automated cricket video segmentation and retrieval system that enables player-wise filtering, event-wise navigation, timestamp-based player-event association, batsman and bowler change tracking, and semantic search of cricket video content.

Specific Objectives

- To preprocess cricket match videos and extract relevant visual regions from broadcast footage for scoreboard and umpire analysis.

- To extract contextual scoreboard text using OCR and normalize the extracted content for more reliable downstream processing.
- To detect major cricket events such as FOUR, SIX, and WICKET using a two-stage umpire-signal recognition pipeline.
- To associate detected events with players using Named Entity Recognition, fuzzy roster matching, and timestamp synchronization.
- To enrich event outputs with ordinal labeling, wicket attribution, over segmentation, and transcript-assisted exploration.
- To implement a web-based platform that supports player-wise filtering, event navigation, and semantic retrieval through MiniLM embeddings and FAISS indexing.
- To evaluate the final system using dataset statistics, classification performance, and practical usability considerations.

2. METHODOLOGY

The Software Development Life Cycle (SDLC) was used to guide the development of the automated cricket video segmentation and retrieval system in a structured and efficient manner. The main purpose of applying the SDLC was to reduce development complexity, improve the quality of the final system, and ensure that the completed solution addressed the practical problem of retrieving player-specific and event-based moments from full-length cricket videos. A systematic development approach was important because the research involved multiple connected components, including video preprocessing, OCR, event detection, player identification, semantic retrieval, and web-based interaction.

The development process began with studying the limitations of existing cricket video analysis approaches and identifying the gaps that remained in full-length match retrieval. Existing methods were useful for event detection and highlight generation, but they did not provide a complete player-centered retrieval experience with structured timestamp-based navigation. Based on this analysis, the requirements of the proposed system were identified. These requirements included the ability to process cricket broadcast videos, extract scoreboard text, detect umpire-based events, associate players with events, generate structured event records, and provide search and navigation support through a web-based interface.

After requirement identification, the development moved through the main SDLC phases of analysis, planning, design, implementation, testing, and deployment-oriented integration. During the analysis and planning stages, the overall workflow of the system was defined and the major processing modules were separated into manageable components. During the design stage, the architecture of the system was organized into stages such as video preprocessing, scoreboard OCR, event detection using umpire signals, player identification, timestamp synchronization, semantic indexing, and web-

based retrieval. This modular design made the implementation process more manageable and reduced the difficulty of integrating multiple machine learning and software components into a single platform.

The implementation stage was carried out mainly in Python using FastAPI as the backend framework. OpenCV, MoviePy, PyTorch, EasyOCR, transformers, Whisper, SentenceTransformer MiniLM, and FAISS were integrated to support the major research functions. Since the project combined several technologies and learned models, the SDLC approach helped reduce development risks by ensuring that each module could be implemented and validated step by step rather than attempting to build the full system all at once. This was especially useful in handling issues such as OCR cleaning, event classification logic, player association, model loading, and output retrieval.

Testing was treated as an important part of the SDLC because the reliability of the final system depended on both individual module behavior and end-to-end pipeline integration. Module-level testing was used to validate OCR extraction, umpire detection, pose classification, and player association, while end-to-end testing was used to check whether the entire system could process a cricket video and generate structured outputs for retrieval. By incorporating testing throughout development, the system could be refined before final integration, reducing the need for major corrections after implementation.

Therefore, the SDLC provided a disciplined framework for developing the proposed cricket video segmentation and retrieval system. It supported the transition from problem identification and requirement analysis to implementation, testing, and final system integration. As a result, the completed software solution was able to function not just as a set of isolated models, but as a coherent web-based system for structured cricket video understanding and retrieval.



Figure 2. 1-Stages of Software Development Life Cycle

2.1 Overall System Architecture

The completed system is designed as an end-to-end multimodal pipeline that processes full-length cricket broadcast videos and converts them into structured, searchable, and player-aware match segments. The overall framework combines computer vision, OCR, player identification, timestamp synchronization, semantic retrieval, and web-based interaction. The purpose of this architecture is not only to detect isolated match events, but also to connect those events to the relevant players and expose the results through an interface that supports practical exploration and retrieval. This design is particularly important in cricket, where broadcast footage contains multiple contextual cues such as scoreboards, umpire gestures, player references, and event timing.

At a high level, the workflow begins when a user uploads a full-length cricket video to the system. After upload, the backend stores the video and begins preprocessing it for further analysis. The video is then sampled frame by frame at fixed intervals so that the system can analyze the scoreboard region and the central visual field without having to process every single frame in the recording. This reduces computational cost while preserving enough contextual information for reliable event detection and player association. The extracted frames are passed through two major processing streams. The first stream focuses on scoreboard analysis and player context extraction through OCR and text post-processing. The second stream focuses on visual event recognition through umpire detection and umpire pose classification. The outputs from these streams are then synchronized using timestamps and converted into structured event records. Those event records are further enriched with player information, ordinal context, and semantic indexing before being exposed through the web interface.

Compared with the initial proposal direction, the completed architecture places stronger emphasis on event detection through umpire signals while still retaining scoreboard-driven player association and transcript-assisted retrieval. This adjustment makes the final

implementation more robust because umpire gestures act as highly structured visual indicators of important cricket events such as FOUR, SIX, and WICKET. At the same time, scoreboard OCR remains essential because it provides batsman, bowler, score, and contextual match information that visual event recognition alone cannot reliably provide. The final architecture therefore treats the cricket broadcast as a multimodal source of information rather than relying on one input source in isolation.

The architecture can be understood through the following sequential stages:

1. Video input and storage
2. Frame extraction and preprocessing
3. Scoreboard OCR and context cleaning
4. Umpire presence detection
5. Umpire pose-based event classification
6. Player identification using OCR text, NER, and roster-aware matching
7. Timestamp synchronization and structured event generation
8. Post-processing and context enrichment
9. Semantic indexing and similarity search
10. Web-based retrieval and interaction

This layered architecture is beneficial because it separates responsibilities across modules. If one module needs improvement, such as OCR cleaning or event classification, it can be refined without fully redesigning the rest of the system. This modular structure also improves maintainability, testing, and extensibility in future research or product development.

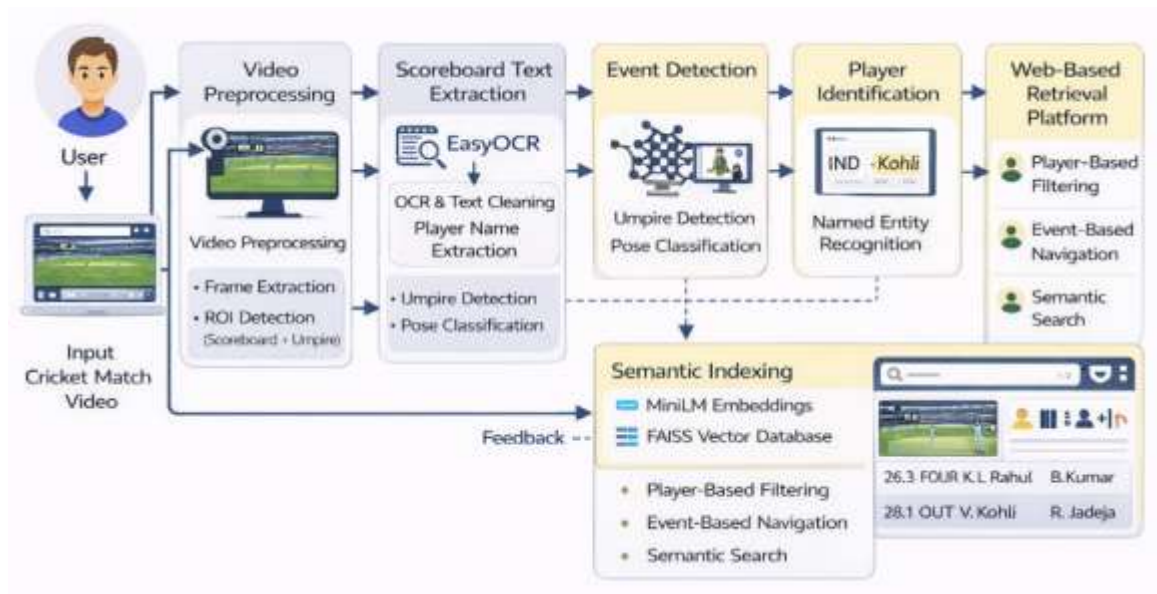


Figure 2. 2-High Level Architecture

Technology Stack

Category	Technologies / Libraries
Programming and backend	Python, FastAPI, JSON, static file serving
Computer vision and video	OpenCV, MoviePy, PIL
Deep learning	PyTorch, timm, torchvision
OCR and language processing	EasyOCR, transformers, AutoTokenizer, Auto Model For Token Classification
Audio and transcript support	Whisper
Semantic retrieval	SentenceTransformer MiniLM, FAISS
Data handling	NumPy, pandas

Table 2. 1-Technology Stack

2.2 Video Preprocessing

The input to the system is a full-length cricket match video obtained from broadcast recordings. The video is processed to extract frames at fixed time intervals. This reduces computational complexity while preserving the visual information needed for event detection and text recognition. Each frame is then analyzed to identify regions containing the scoreboard and the umpire area. This preprocessing stage helps the later modules focus only on relevant visual regions, improving both detection accuracy and efficiency

2.3 Scoreboard Text Extraction

Scoreboard overlays in cricket broadcasts contain useful contextual information such as player names, scores, and match statistics. In this work, scoreboard text is extracted from video frames using EasyOCR, a deep-learning-based optical character recognition framework. The extracted text is cleaned and processed to identify player names and relevant match context. Since OCR outputs may contain noise due to broadcast graphics, camera movement, and overlay variation, text normalization is applied to improve reliability. The extracted text is then used to estimate the batsmen and bowler present at a given moment in the match.

2.4 Event Detection Using Umpire Signals

Key cricket events such as FOUR, SIX, and WICKET are detected using visual cues from umpire signals. The proposed event detection pipeline uses two deep learning models. First, an EfficientNet-B0 classifier is used to detect the presence of an umpire in a video frame. Once an umpire is detected, a second model based on ResNet-18 classifies the specific pose corresponding to the event type. This two stage design improves detection accuracy by separating umpire detection from pose-based event classification. Detected events are recorded together with their timestamps to support accurate video segmentation

```

events.json
C:\Users\dinal> OneDrive\ Desktop\ PlayerBasedSegmentation\ backend\ cricket_system\ matches\ 0be14c266d814b6eb613baefc2a15e789_M46_0\ events.json
24 {
25   "event_type": "SIX",
26   "timestamp_sec": 53.4,
27   "timestamp_str": "00:00:53.400",
28   "start_sec": 53.199999999999996,
29   "end_sec": 55.0,
30   "confidence": 0.9976153373718262,
31   "source": "umpire_signal",
32   "det_p_umpire": 0.9999997615814209,
33   "pose_label": "six",
34   "pose_conf": 0.9976153373718262,
35   "crop_used": "right",
36   "debug_cluster_size": 10,
37   "debug_majority_count": 10,
38   "batsman_1": "ABHISHEK",
39   "batsman_2": "GILL",
40   "bowler": "KPH",
41   "commentary": "",
42   "event_type_normalized": "SIX",
43   "type_index": 2,
44   "type_total": 2
45 },
46 {
47   "event_type": "FOUR",
48   "timestamp_sec": 67.4,
49   "timestamp_str": "00:01:07.400",
50   "start_sec": 66.0,
51   "end_sec": 67.4,
52   "confidence": 0.998913849697876,
53   "source": "umpire_signal",
54   "det_p_umpire": 0.9999984502792358,
55   "pose_label": "four",
56   "pose_conf": 0.998913849697876,
57   "crop_used": "left",
58   "debug_cluster_size": 8,
59   "debug_majority_count": 8,
60   "batsman_1": "ABHISHEK",

```

Figure 2. 3-Event Detection Example

2.5 Player Identification and Post-processing

After extracting textual information and detecting events, the system associates players with the detected match events. Player identification is performed using Named Entity Recognition (NER) applied to the OCR-extracted text. The NER model identifies player entities from the text and maps them to the corresponding batsmen and bowler present in the frame. By combining OCR results with NER-based entity recognition, the system can determine which players are involved in specific match events more reliably.

```

29  },
30  {
31    "player_name": "SHORIFUL",
32    "role": "batsman_2",
33    "first_seen_sec": 18.0,
34    "last_seen_sec": 32.0,
35    "num_occurrences": 15
36  },
37  {
38    "player_name": "TASKIN",
39    "role": "batsman_1",
40    "first_seen_sec": 18.0,
41    "last_seen_sec": 32.0,
42    "num_occurrences": 15
43  },
44  {
45    "player_name": "ABHISHEK",
46    "role": "batsman_1",
47    "first_seen_sec": 36.0,
48    "last_seen_sec": 67.0,
49    "num_occurrences": 32
50  },
51  {
52    "player_name": "GILL",
53    "role": "batsman_2",
54    "first_seen_sec": 36.0,
55    "last_seen_sec": 67.0,
56    "num_occurrences": 32
57  },
58  {
59    "player_name": "NAZUM",
60    "role": "bowler",
61    "first_seen_sec": 36.0,
62    "last_seen_sec": 56.0,
63    "num_occurrences": 11
64  },
65  }

```

Figure 2. 4-Player Identification Example

2.6 Semantic Indexing and Retrieval

To enable efficient retrieval of video segments, the extracted event data and textual descriptions are converted into vector representations using MiniLM sentence embeddings. These embeddings capture semantic relationships between textual queries and video segments. The embeddings are indexed using FAISS, a high-performance similarity search library that supports fast nearest-neighbor search in large embedding spaces. This allows users to perform semantic queries such as searching for specific players or events.

2.7 Web-based Platform

The final component of the system is a web-based platform that allows users to interact with indexed cricket videos. The platform provides functions such as player-based filtering, event-based navigation, and semantic search. Users can quickly retrieve relevant match segments by selecting players, events, or search queries. The platform displays timestamped video segments, allowing efficient navigation through full length cricket matches.

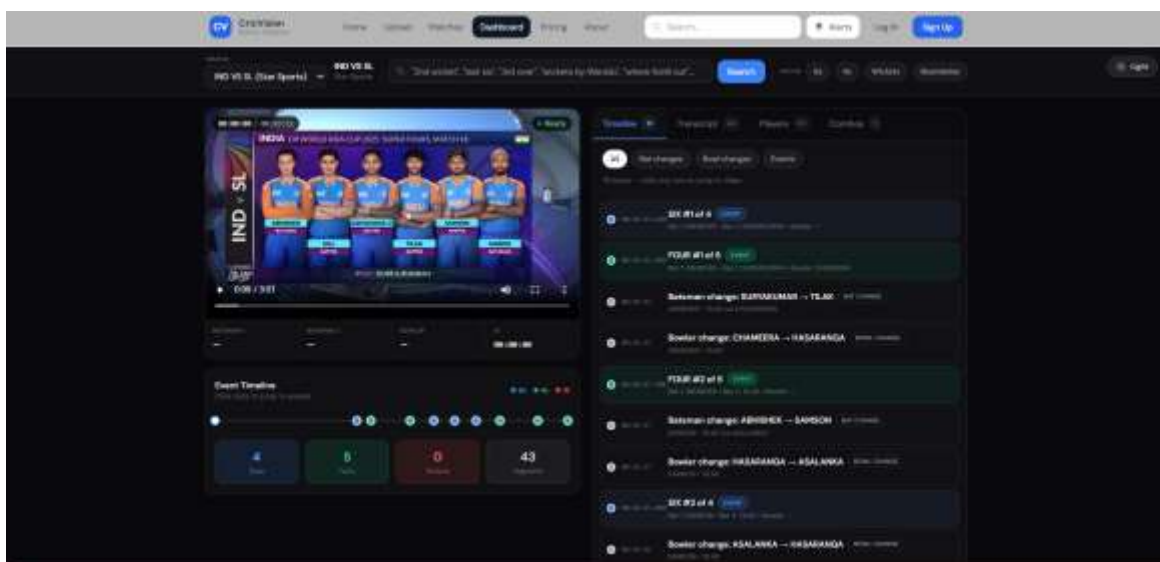


Figure 2. 5-Image of Website

2.8 Commercialization Aspect of the Product

The completed system has practical commercialization potential in sports analytics, coaching, and fan-engagement domains. Cricket academies and coaching staffs can use the platform to retrieve player-specific moments without reviewing complete match recordings manually. Broadcasters and digital sports platforms can use structured event indexing to produce better highlight experiences and searchable archives. Cricket content creators and analysts can also benefit from timestamped event retrieval for faster breakdowns and reports.

From a product perspective, the system can be offered as a web-based software platform or analytics service. Possible commercial models include subscription access for academies, match-processing services for clubs and analysts, or platform licensing for cricket media and training providers. The semantic retrieval layer and player-event association provide additional value beyond ordinary highlight tools because they support targeted exploration instead of only generic summaries.

Plan	Target Users	Monthly Price	Video Upload Limit
Starter	Individual coaches, small creators, students	\$29 / month	5 videos / month
Professional (Pro)	Cricket academies, analysts, small teams	\$99 / month	25 videos / month
Enterprise	Large organizations, broadcasters, sports platforms	Custom Pricing	Unlimited videos

Table 2. 2-Subscription Plan

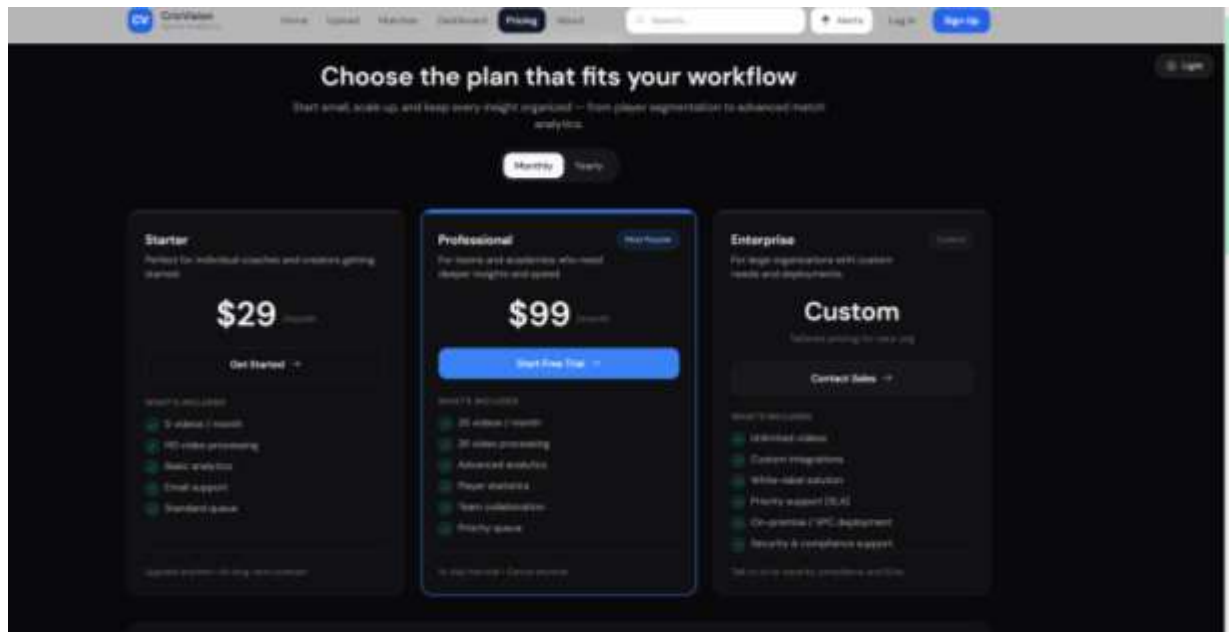


Figure 2. 6-Image of Subscription Plan

2.9 Testing and Implementation

The backend implementation is written primarily in Python. Video processing, OCR, deep learning inference, transcript extraction, and retrieval are integrated within a FastAPI-based application. Important implementation considerations include robust checkpoint loading for trained models, device selection between CPU and GPU, modular cache management for loaded models, and separate storage locations for uploaded videos, processed match data, and generated outputs.

The implementation includes utility functions for OCR cleaning, player candidate generation, fuzzy roster matching, NER inference, score parsing, event clustering, transcript association, wicket attribution, and over segmentation. This modular structure supports easier testing and future improvement.

System testing was conducted at the module level and at the end-to-end level. Module-level testing focused on OCR extraction behavior, event classification reliability, and correctness of player association logic. End-to-end testing evaluated whether the full pipeline could process a cricket video, generate structured event outputs, and make those outputs retrievable through the web interface.

Examples of system-level testing scenarios carried out for the completed framework are described below.

Field	Description
Test case ID	TC01
Test Case Scenario	Process a full-length cricket broadcast video and generate structured event records.
Test input data	Uploaded cricket match video.
Test procedure	1. Upload a cricket video to the platform. 2. Allow the backend to extract frames at fixed intervals. 3. Apply scoreboard OCR and text cleaning. 4. Apply umpire detection and pose classification. 5. Synchronize detected events with timestamps. 6. Generate structured event records.
Expected output	The system should produce timestamped event records containing detected events and associated player-related context.
Actual output	The system generated structured event records with timestamps and contextual information.
Test result	Pass

Table 2. 3-Generate event records from cricket video test case

Field	Description
Test case ID	TC02
Test Case Scenario	Detect a cricket event from umpire gestures in broadcast frames.
Test input data	Extracted video frames containing umpire gestures.
Test procedure	1. Pass extracted frames to the umpire detection model. 2. Filter relevant frames with sufficient confidence. 3. Send valid frames to the pose classifier. 4. Assign the correct event label.
Expected output	The system should classify the relevant umpire gesture into the appropriate event category such as FOUR, SIX, or WICKET.
Actual output	The system classified umpire gestures into event categories and generated corresponding event records.
Test result	Pass

Table 2. 4-Detect umpire-based cricket event test case

Field	Description
Test case ID	TC03
Test Case Scenario	Associate batsman and bowler information with detected event timestamps.
Test input data	OCR-extracted scoreboard text, NER outputs, player rosters, and event timestamps.
Test procedure	1. Extract scoreboard text using OCR. 2. Clean the text and generate candidate player tokens. 3. Apply NER and fuzzy roster matching. 4. Attach nearest player predictions to detected events using timestamps.
Expected output	The system should assign relevant batsman and bowler information to event records.
Actual output	Event records were generated with associated player information based on OCR text and matching logic.
Test result	Pass

Table 2. 5-Associate players with detected events test case

Field	Description
Test case ID	TC04
Test Case Scenario	Retrieve processed cricket moments through semantic search.
Test input data	Indexed event descriptions and a user search query.
Test procedure	1. Convert event descriptions into MiniLM embeddings. 2. Store embeddings in the FAISS index. 3. Submit a search query through the platform. 4. Compare the query embedding against indexed events.
Expected output	The system should return relevant match events and timestamps related to the query.
Actual output	The system returned contextually relevant match moments through semantic retrieval.
Test result	Pass

Table 2. 6-Semantic retrieval of processed cricket moments test case

Functional Testing

Functional testing was conducted to confirm that each major feature of the cricket video segmentation platform worked according to its intended purpose. Since the completed system was developed as a retrieval-oriented web platform rather than only a standalone model pipeline, functional testing covered both backend processing behavior and user-facing operations.

The major functional areas tested were as follows:

- Video upload and processing – The system was tested to confirm that users could upload cricket videos and initiate backend processing successfully.
- Scoreboard OCR extraction – The OCR module was tested to ensure that scoreboard text could be extracted and cleaned from broadcast frames.
- Event detection – The visual pipeline was tested to verify that umpire presence could be detected and that umpire poses could be classified into the required cricket event categories.
- Player identification – The OCR, NER, and fuzzy matching stages were tested together to ensure meaningful batsman and bowler assignment.
- Timestamp synchronization – The system was tested to confirm that detected events and player associations could be mapped correctly to video timestamps.
- Structured event generation – The output generation process was tested to ensure that final records contained event type, timestamp, and contextual player information.
- Semantic retrieval – The semantic indexing layer was tested to confirm that processed match moments could be searched and retrieved through query-based interaction.

- Web-based result exploration – The platform interface was tested to ensure that users could browse, filter, and retrieve relevant match outputs through the web application.

Functional testing therefore confirmed that the completed framework operated correctly as an integrated cricket video segmentation and retrieval system.

Non-functional Testing

Non-functional testing was conducted to evaluate the broader quality characteristics of the system beyond output correctness. For this study, the most relevant non-functional aspects were performance, stability, usability, and compatibility of the final retrieval workflow.

- Usability testing – Usability testing was performed to ensure that the processed outputs could be explored in a practical and understandable manner through the web-based platform. Special attention was given to the presentation of timestamped event records, player-based filtering, event-based navigation, and semantic search so that users could interact with the final outputs efficiently without needing to manually inspect the entire video.
- Performance testing – Performance testing focused on confirming that the system could process cricket videos in a practical way without unnecessary delays in the major stages of frame extraction, OCR, event detection, and retrieval. The use of fixed-interval frame sampling and the two-stage event-detection pipeline helped reduce computational cost while maintaining meaningful outputs.
- Stability testing – Stability testing was carried out to ensure reliable backend behavior during repeated processing sessions. The implementation included robust checkpoint loading, cache-based model reuse, structured storage management, and controlled timestamp handling, all of which contributed to more stable execution across different runs.

- Compatibility testing – Compatibility testing was considered in relation to the web-based access model of the platform, ensuring that the retrieval interface remained usable for interacting with processed cricket video outputs in a browser-based environment.

Overall, non-functional testing showed that the completed system was not only technically functional, but also practical as a usable cricket video segmentation and retrieval platform.

3. RESULTS AND DISCUSSION

3.1 Dataset Summary

Three primary datasets were used in the completed research: an umpire versus non-umpire image dataset, an umpire pose dataset, and a labeled scoreboard OCR dataset. These datasets were used to support the visual and textual components of the system.

1.Umpire Detection Dataset

Dataset Split	Umpire	Non-Umpire	Total
Training Set	446	638	1084
Test Set	111	159	270
Total	557	797	1354

Table 3. 1-Umpire Detection Dataset

2.Umpire Pose Classification Dataset

Dataset Split	FOUR	SIX	OUT	No-Action	Total
Training Set	130	151	202	162	645
Test Set	32	38	51	40	161
Total	162	189	253	202	806

Table 3. 2-Umpire Pose Classification Dataset

3.Labeled Scoreboard OCR Dataset

The labeled scoreboard OCR dataset contained 3,800 samples and supported the extraction of player-related text from scoreboard overlays. Together, these datasets supported the multimodal design of the proposed framework.

3.2 Umpire Detection Performance

The first-stage umpire detection model achieved an overall test accuracy of 0.91. The non-umpire class achieved precision 0.95, recall 0.90, and F1-score 0.93, while the umpire class achieved precision 0.87, recall 0.94, and F1-score 0.90. The weighted F1-score was 0.92.

These results show that the filtering stage is effective in isolating candidate frames before event classification. This is important because reducing irrelevant frames directly improves the quality of downstream event recognition.

Classification Report:				
	precision	recall	f1-score	support
non_umpire	0.95	0.90	0.93	159
umpire	0.87	0.94	0.90	111
accuracy			0.91	270
macro avg	0.91	0.92	0.91	270
weighted avg	0.92	0.91	0.92	270

Figure 3. 1-Classification Report of Umpire Detection Model

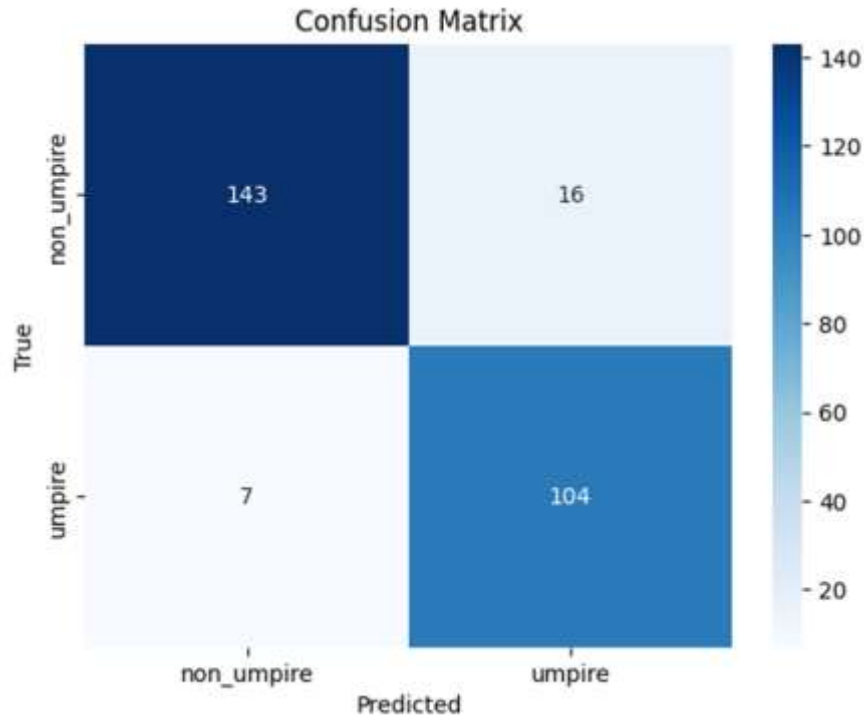


Figure 3. 2-Confusion Matrix of Umpire Detection Model

3.3 Event Classification Performance

The ResNet-18 umpire pose classifier achieved an overall test accuracy of 0.83 on four classes: FOUR, SIX, OUT, and NO-ACTION.

Class	Precision	Recall	F1-score
FOUR	0.90	0.84	0.87
NO-ACTION	0.83	0.85	0.84
OUT	0.83	0.78	0.81
SIX	0.79	0.87	0.82

Table 3. 3-Event Classification Performance

Performance was reasonably balanced across the classes, although misclassifications still appeared when gestures were visually similar or when camera angles were non-standard. Confusions were especially likely between OUT and SIX or between FOUR and NO-ACTION.

Even with these challenges, the model demonstrated that cricket umpire gestures can serve as a reliable basis for event detection when combined with high-confidence gating and temporal clustering.

Classification Report:				
	precision	recall	f1-score	support
four	0.90	0.84	0.87	32
no_action	0.83	0.85	0.84	40
out	0.83	0.78	0.81	51
six	0.79	0.87	0.82	38
accuracy			0.83	161
macro avg	0.84	0.84	0.84	161
weighted avg	0.83	0.83	0.83	161

Figure 3. 3-Classification Report of Event Classification Model

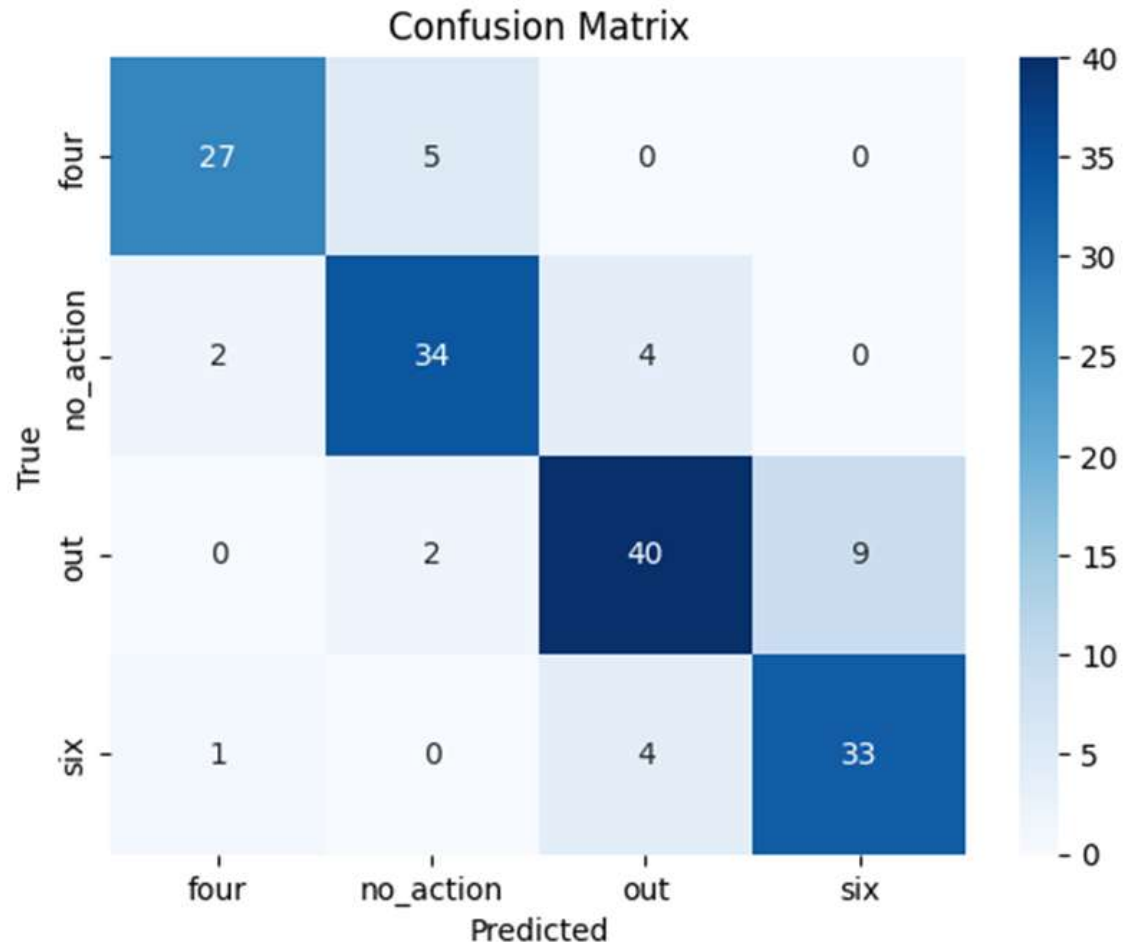


Figure 3. 4-Confusion Matrix of Event Classification Model

3.4 OCR, Player Context, and Event Association

Scoreboard OCR provided direct contextual information that was difficult to infer reliably from raw player appearance alone. The extracted text contributed player names, score information, and match state cues that helped determine batsmen and bowler around each detected event.

The combination of OCR token extraction, NER, and fuzzy roster matching improved the robustness of player association. This made it possible to transform isolated event detections into structured event records that included event type, event timestamp, batsman information, and bowler information.

Additional post-processing further improved usefulness. Ordinal event enrichment, wicket attribution, and over segmentation produced a richer representation of the match that can support downstream search and exploration.

3.5 Practical Usability of the Final Platform

The final platform supports several user-facing capabilities: player-wise filtering, event-based navigation, over-based browsing, transcript-assisted exploration, wicket attribution display, and semantic search with direct video playback links. These features demonstrate that the project was completed not only as a classification study but also as an interactive retrieval system.

The system therefore addresses an important practical gap in cricket video analysis by allowing end users to move directly from an uploaded match video to structured, searchable, and timestamped outputs.

3.6 Discussion

The completed research shows that a multimodal approach is effective for cricket video understanding. The strongest aspect of the system is that it does not rely on a single signal. Instead, it combines scoreboard OCR, event recognition through umpire gestures, player identification, timestamp synchronization, transcript assistance, and semantic retrieval.

The event detection results show that the chosen two-stage design is a sensible architectural decision. The first classifier removes irrelevant frames, while the second classifier focuses on event categories. This decomposition improves robustness and reduces unnecessary false positives in the later stage.

The OCR and player identification pipeline provides contextual grounding for the detected events. Without this step, event predictions would remain isolated and less useful. By attaching players to timestamps and then enriching the records with wicket attribution and over segmentation, the system becomes more valuable to coaches, analysts, and fans.

The study also reveals several limitations. OCR quality can degrade under poor broadcast graphics, motion blur, and low-resolution overlays. Pose-based classification can be affected by occlusion, unusual camera framing, and visually ambiguous gestures. The semantic retrieval quality also depends on the quality of the generated textual descriptions and indexed context.

3.7 Summary of Student Contribution

This draft final report is presented as the completed research work of the candidate. The primary student contribution included literature analysis, problem formulation, dataset preparation, model pipeline design, backend implementation, OCR processing, event detection integration, player association logic, semantic retrieval integration, web platform assembly, experimentation, result analysis, and report preparation.

Student Name	Student ID	Contribution Summary
R M N D Rathnayake	IT22141606	End-to-end research design, implementation, testing, result analysis, and documentation of the completed system.

4. CONCLUSION AND RECOMMENDATIONS

This research presented an automated cricket video segmentation and retrieval framework for identifying player-related and event-based moments in full-length broadcast cricket videos. The study was motivated by the practical difficulty of manually reviewing lengthy match recordings to locate specific events or player performances. To address this limitation, the proposed system combined scoreboard text extraction, visual event detection, player association, timestamp synchronization, and semantic retrieval within a single operational pipeline.

The implemented framework used EasyOCR to extract scoreboard information from broadcast frames and recover useful contextual details such as player names and match-related text. Key match events were detected through a two-stage umpire-signal recognition pipeline, where EfficientNet-B0 was used for umpire detection and ResNet-18 was used for pose-based event classification. In addition, player identification was improved through OCR-based text processing, Named Entity Recognition, and roster-aware matching, enabling detected events to be linked with the relevant batsmen and bowler. These outputs were synchronized using timestamps to generate structured event records and meaningful video segments.

A major contribution of the study is that it moved beyond isolated event detection and developed a retrieval-oriented system for cricket video exploration. By converting event descriptions and contextual outputs into MiniLM sentence embeddings and indexing them with FAISS, the system supported semantic search and efficient retrieval of relevant match moments. The web-based platform further improved practical usability by allowing users to interact with processed videos through player-wise filtering, event-based navigation, and searchable timestamped outputs.

The results demonstrate that the proposed multimodal approach can reduce manual effort in cricket video analysis and improve access to important player-specific and event-based moments. The combination of OCR-based contextual extraction, umpire-signal-based event recognition, player association, and semantic retrieval proved to be a practical and

effective strategy for understanding cricket broadcast videos. The study therefore contributes a structured and technically sound approach for transforming unorganized cricket match recordings into searchable multimedia content.

Although the framework achieved useful results, certain limitations remain. OCR performance can still be affected by noisy scoreboard overlays, motion blur, and broadcast quality variations. Similarly, pose-based event classification may be influenced by occlusions, non-standard camera views, and visually similar umpire gestures. Despite these challenges, the completed system provides a strong foundation for cricket video intelligence and retrieval.

In conclusion, this research successfully developed a practical automated framework for cricket video segmentation by player identity and event type. The study demonstrates how multimodal AI techniques can be integrated into a usable retrieval platform that supports analysts, coaches, and fans in exploring cricket match content more efficiently. Future improvements can focus on increasing OCR robustness, expanding the pose dataset, and strengthening temporal modeling for even more accurate event detection and player-event association.

Recommendations and Future Work

- Improve OCR robustness through stronger scoreboard localization, super-resolution, and domain-specific text correction.
- Expand the event taxonomy beyond FOUR, SIX, and WICKET to include wides, no-balls, reviews, and near-miss moments.
- Use larger and more diverse umpire gesture datasets to improve generalization across broadcast styles and camera angles.
- Introduce temporal deep learning models that reason across frame sequences rather than isolated frame samples.
- Automation of scoring system

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APPENDICES

Appendix A: Images Of Website



